Data link layer

Introduction to the Data Link Layer

Definition and Overview

The data link layer is responsible for the node-to-node delivery of the message. It acts as the bridge between raw hardware signals and structured data, ensuring reliable communication between directly connected devices.

- When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address.
- Packet in the Data Link layer is referred to as Frame. Switches and Bridges are common Data Link Layer devices.

Introduction to the Data Link Layer

- The packet received from the Network layer is further divided into frames depending on the frame size of the NIC (Network Interface Card). DLL also encapsulates Sender and Receiver's MAC address in the header.
- The Receiver's MAC address is obtained by placing an ARP (Address Resolution Protocol) request onto the wire asking, "Who has that IP address?" and the destination host will reply with its MAC address.

Introduction to the Data Link Layer

Role and Importance

- Ensures node-to-node delivery of data within a local network
- Guarantees error-free transmission, forming the reliability foundation for all upper layers
- Without it, raw signals would have no structure, making organized communication impossible

Functions of the Data Link Layer

1. **Framing** - Encapsulates packets into frames with clear start/end boundaries for proper identification

2. **Physical Addressing** - Assigns unique 48-bit MAC addresses to devices for accurate frame delivery

3. **Flow Control** - Matches transmission speeds between sender and receiver to prevent data overflow

4. **Error Control** - Detects and corrects flipped bits using techniques like CRC and parity bits; notifies sender of errors

5. **Access Control** - Coordinates shared media access using protocols like CSMA/CD to prevent data collisions

Sub-Layers of the Data Link Layer

1. **Logical Link Control (LLC)** - The upper sub-layer of the Data Link Layer that communicates with the Network Layer. Functions

-Manages flow of data between applications and network services.

-Provides error notifications and acknowledgments.

-Allows multiple protocols to use the same network interface (multiplexing).

Sub-Layers of the Data Link Layer

2. **Media Access Control (MAC)** - The lower sub-layer that controls how devices access and use the network medium.

Functions - Handles MAC addressing for identifying devices.

-Controls access to the physical network (who sends data first).

-Converts packets into frames and sends them to the Physical Layer.

Example: When multiple computers share the same network cable or Wi-Fi, MAC decides which device can transmit data.

Sub-Layers of the Data Link Layer

3. **MAC Address** - A MAC (Media Access Control) address is a unique hardware address assigned to a network device for identification in a local network.

Function

-Identifies devices in the Data Link Layer.

-Helps send data frames to the correct device in a network.

-Example 00:1A:2B:3C:4D:5E (a typical 48-bit hexadecimal MAC address)

Sub-Layers of the Data Link Layer

4. **Framing** - is the process of dividing data into smaller units called frames for transmission over a network.

Functions:

-Adds source and destination MAC addresses to identify sender and receiver.

-Separates one message from another during transmission.

-Helps the receiver recognize the start and end of a data frame.

Methods of Framing :

1. **Character Count** - A framing method where the header contains a number that tells the total length of the frame.
2. **Character Stuffing** - is a framing method where an extra byte (ESC – escape character) is added when special flag bytes appear in the data.
3. **Bit Stuffing** - a framing method where extra bits are inserted into the data stream.
4. **Physical Layer Coding Violations** - A framing method that uses special signal patterns that are not normally used in encoding.

Error Detection and Correction

When data is transmitted over a network, it doesn't always arrive perfectly. Noise, interference, and signal degradation can cause bits to get flipped or lost along the way — meaning what the sender sends isn't always what the receiver gets.

Error Detection is the process of identifying whether any errors occurred during transmission.

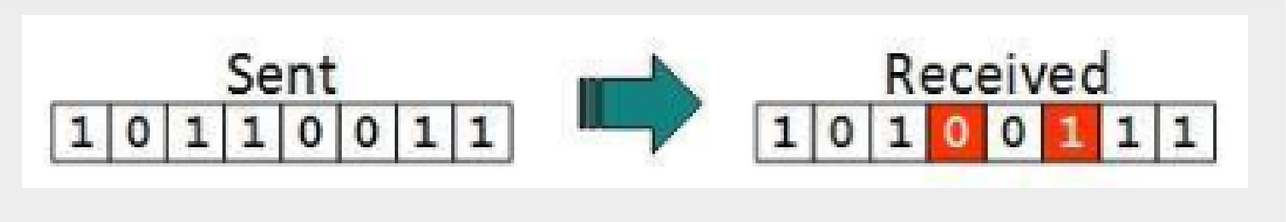
Error Correction goes a step furtherit not only finds the error but also fixes it, without needing to request the data again.

These mechanisms are essential to making sure data arrives accurately and reliably across any network

TYPES OF ERROR

Single-bit Error — Only one bit in a data frame gets corrupted during transmission. The rest of the data remains intact.

Multiple-bit Error — More than one bit in a frame gets corrupted, but they are not necessarily next to each other. This makes it harder to detect compared to a single-bit error.



Burst Error — Two or more consecutive bits in a frame are corrupted. This is the most common type in real-world transmission and the most challenging to handle.

ERROR DECTECTION METHODS

Parity Check A single extra bit is added to the data. The sender counts the number of 1s and adds a bit to make that count either always even (even parity) or always odd (odd parity). The receiver checks this count,if it doesn't match, an error is detected. Works well for single-bit errors but struggles with multiple errors.

CRC (Cyclic Redundancy Check) The sender divides the data bits using a polynomial-based divisor and attaches the remainder to the data, this combined piece is called a codeword. The receiver does the same division; if the remainder is zero, the data is clean. If not, corruption occurred. More reliable than parity.

Checksum — Is a common method where data is divided into segments, summed up, and the total is sent along. The receiver does the same sum and checks if they match.

ERROR DECTECTION METHODS

Error correction (Hamming Code)

The page explains the concept behind Hamming Code:

- To correct an error, the receiver needs to know exactly which bit is wrong, not just that an error exists.
- Extra redundant bits (parity bits) are added at specific positions in the data.
- For m data bits, you need r redundant bits such that $2^r \geq m + r + 1$. This gives enough combinations to pinpoint the exact faulty bit location.
- This is the foundation of Hamming Code, it uses these redundant bits to both detect and correct single-bit errors automatically.

Flow Control

When two devices communicate over a network, they don't always operate at the same speed. A fast sender can easily overwhelm a slow receiver by sending too much data at once, causing data loss. Flow Control is a set of techniques that manages how much data a sender can transmit before needing a response from the receiver keeping communication smooth and balanced.

ERROR DECTECTION METHODS

MAIN TECNIQUES/ PROTOCOLS used in flow control

Stop-and-Wait - The sender sends one frame and then stops and waits for an acknowledgment from the receiver. Once the acknowledgment is received, the sender knows the receiver is ready and sends the next frame.

Downside: Only one frame is sent at a time, which wastes time and makes it inefficient for large data transfers.

Sliding Window - This protocol improves on Stop-and-Wait by allowing multiple frames to be sent at once before waiting for an acknowledgment. Both the sender and receiver have buffers called windows, and the sender can keep transmitting frames until its window is full. Once acknowledgments are received, the window advances and more frames are sent.

Advantage: Much more efficient than Stop-and-Wait since it doesn't waste time sitting idle between each frame.

Media Access Control Protocols

1. Random Access No scheduling, any station transmits when ready. Collisions are possible.

ALOHA - Send anytime. If no ACK received, wait a random time and retry.

Pure ALOHA: 18.4% efficiency

Slotted ALOHA: 36.8% efficiency (transmit only at slot boundaries) CSMA/CD (Wired Ethernet) - Listen before sending. If collision detected while transmitting, stop, send a jam signal, wait random time, retry. CSMA/CA (Wi-Fi) - Listen before sending. Wait an IFS period + random backoff to avoid collisions. Uses ACKs to confirm delivery. Optional RTS/CTS for hidden node problem.

Media Access Control Protocols

2. Controlled Access Transmission is coordinated - no collisions possible.

Polling - A master station asks each slave "do you have data?" one by one. Only polled stations may transmit.

Token Passing - A token circulates the network. Only the station holding the token may transmit, then it passes the token along.

Media Access Control Protocols

3. Channelization The channel is divided so each station gets its own dedicated resource.

FDMA - Each station gets its own frequency band. Used in FM radio, 1G cellular.

TDMA - Each station gets its own time slot. Used in GSM (2G), satellite.

CDMA - All stations transmit simultaneously using unique codes to separate signals. Used in 3G cellular, GPS.

Data Link Layer Technologies

HDLC (High-Level Data Link Control)

- Used when two devices are directly connected
- Organizes data into frames
- Detects errors and ensures reliable transmission

PPP (Point-to-Point Protocol)

- Similar to HDLC but more secure
- Verifies user identity
- Works with different network systems
- Used in older dial-up connections

Ethernet (IEEE 802.3)

- Most common technology in local networks
- Uses MAC addresses
- Sends data in frames
- Connects many devices in one network

Switching and Network Organization

Switching (Data Link Layer)

- Sends data to the correct device in a network

Bridge vs Switch

Bridge

- Connects two network segments
- Reduces unnecessary traffic

Switch

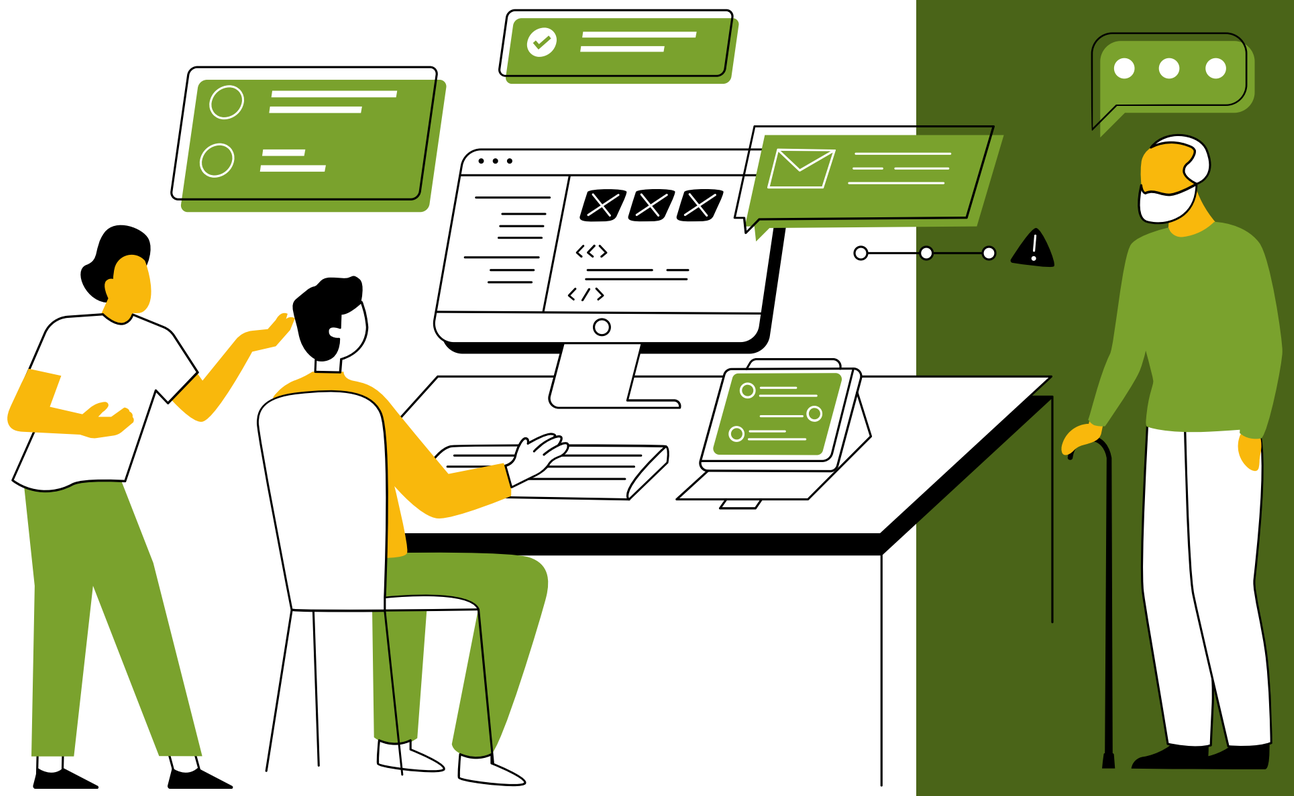
- More advanced than a bridge
- Connects many devices
- Sends data faster and more efficiently

Summary

The Data Link Layer, the seco layer of the OSI Model, mainly ensures reliable and organized data transfer between devices on the same network. -It receives data from the higher layer and divides it into frames. --th-en add addressing information. -The layer also controls how devices access the communication medium, and checks for errors during transmission. -If errors are detected, it can request retransmission to maintain data accuracy. Through these processes, the Data Link Layer acts as the connection between the physical transmission of signals and the logical communication processes of the network.

Conclusion

Overall, the Data Link Layer is essential for maintaining organized and dependable data transmission in a network. By controlling how devices access the medium, structuring data into frames, and detecting transmission errors, it helps ensure that data is delivered accurately and efficiently. This layer acts as the bridge between the physical transmission of signals and higher-level network processes, contributing significantly to the stability and performance of modern communication networks.



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